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SCALE-AWARE SELF-SUPERVISED MONOCULAR DEPTH WITH SPARSE RADAR SUPERVISION

Abstract

Systems and methods are provided for training a depth model to recover scale factor for self-supervised depth estimation in monocular images. According to some embodiments, a method comprises receiving an image representing a scene of an environment; deriving a depth map for the image based on a depth model, the depth map comprising depth values for pixels of the image; estimating a first scale for the image based the depth values; receiving depth data captured by a range sensor, the depth data comprising a point cloud representing the scene of the environment, the point cloud comprising depth measures; determining a second scale for the point cloud based on the depth measures; determining a scale factor based the second scale and the first scale; and updating the depth model based on the scale factor, wherein the depth model generates metrically accurate depth estimates based on the scale factor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims the benefit of U.S. patent application Ser. No. 17/578,830 filed on Jan. 19, 2022, which is hereby incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates generally to systems and methods for depth estimation from images, and in particular, to training a depth model sparse range sensor data to recover scale factor for self-supervised depth estimation in monocular images.

DESCRIPTION OF RELATED ART

[0003] Various devices that operate autonomously or that provide information about a surrounding environment use sensors that facilitate perceiving obstacles and additional aspects of the surrounding environment. For example, a robotic device may use information from the sensors to develop an awareness of the surrounding environment in order to navigate through the environment. In particular, the robotic device uses the perceived information to determine a 3-D structure of the environment in order to identify navigable regions and avoid potential hazards. The ability to perceive distances through estimation of depth using sensor data provides the robotic device with the ability to plan movements through the environment and generally improve situational awareness about the environment.

[0004] However, depending on the available onboard sensors, the robotic device may acquire a limited perspective of the environment, and, thus, may encounter difficulties in distinguishing between aspects of the environment.

[0005] That is, various sensors perceive different aspects of the environment differently and also have different implementation characteristics. For example, a light detection and ranging (LiDAR) sensor is effective at perceiving depth in the surrounding environment but suffers from difficulties such as high costs and can encounter errors in certain weather conditions, while radar sensors suffer from sparsity and noise. Moreover, other sensors, such as stereo cameras, function to effectively capture depth information but also suffer from difficulties with cost, limited field-of-view, and so on. While monocular cameras can be a cost-effective approach, the sensor data from such cameras does not explicitly include depth information. Instead, the robot implements processing routines that derive depth information from the monocular images.

[0006] However, leveraging monocular images to perceive depth can also suffer from difficulties such as limited resolution, image artifacts, difficulties with training the processing routines (e.g., expensive or limited availability of training data), and so on. As such, many difficulties associated with determining depth data persist that may result in reduced situational awareness for a device, and, thus, difficulties in navigating or performing other associated functions. Many approaches to monocular depth estimation estimate up to an unknown scale factor with the scale applied as a post-processing step at, for example, test time. As such, many difficulties associated with determining depth data persist such as scale ambiguities and other difficulties that may result in reduced situational awareness for a device, and, thus, difficulties in navigating or performing other associated functions.

BRIEF SUMMARY OF THE DISCLOSURE

[0007] According to various embodiments of the disclosed technology, systems and methods for training a depth model sparse range sensor data are provided to recover scale factor for self-supervised depth estimation in monocular images.

[0008] In accordance with some embodiments, a method for depth estimation from monocular images

is provided. The methods comprises receiving an image captured by an image sensor, the image comprising pixels representing a scene of an environment; deriving a depth map for the image based on a depth model, the depth map comprising predicted depth values for a plurality of the pixels of the image; estimating a first scale for the image based the plurality of predicted depth values; receiving depth data captured by a range sensor, the depth data comprising a point cloud representing the scene of the environment, the point cloud comprising depth measures for a plurality of points of the point cloud; determining a second scale for the point cloud based on the depth measures; determining a scale factor based on a comparison of the second scale with the first scale; and updating the depth model based on the scale factor, wherein the depth model generates metrically accurate depth estimates based on the scale factor.

[0009] In another aspect, a system is provided that comprises a memory and one or more processors that are configured to execute machine readable instructions stored in the memory for performing a method. The method comprises training a depth model at a first stage according to self-supervised photometric losses generated from at least a first monocular image; determining a single scale factor from a depth map of the first monocular image and a sparse point cloud generated by a range sensor; and training the depth model at a second stage according to a supervised loss based on the single scale factor, wherein the depth model trained according to the second stage generates metrically accurate depth estimates of monocular images based on the single scale factor.

[0010] Other features and aspects of the disclosed technology will become apparent from the following detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the features in accordance with embodiments of the disclosed technology. The summary is not intended to limit the scope of any inventions described herein, which are defined solely by the claims attached hereto.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The figures are provided for purposes of illustration only and merely depict typical or example embodiments.

[0012] FIG. 1 is a schematic representation of an example vehicle with which embodiments of semi-supervised scale-aware learning systems and methods disclosed herein may be implemented.

[0013] FIG. 2 illustrates an example architecture for semi-supervised scale-aware learning in accordance with embodiments of the systems and methods described herein.

[0014] FIG. 3 illustrates a schematic block diagram of an example semi-supervised learning circuit in accordance with embodiments of the systems and methods described herein.

[0015] FIG. 4 illustrates a schematic block diagram of an example depth model according to embodiments disclosed herein

[0016] FIG. 5 illustrates a schematic block diagram of an example pose model according to embodiments disclosed herein.

[0017] FIG. 6 is a diagram illustrating an example architecture for semi-supervised scale-aware learning of a depth model for monocular depth estimation in accordance with various embodiments.

[0018] FIG. 7 is an operational flow diagram illustrating an example process for semi-supervised scale-aware learning of a depth model for monocular depth estimation in accordance with various embodiments.

[0019] FIG. 8 is a flow chart illustrating example operations for implementing semi-supervised scale-aware learning of a depth model for monocular depth estimation in accordance with various embodiments.

[0020] FIG. 9 is an example computing component that may be used to implement various features of embodiments described in the present disclosure.

[0021] The figures are not exhaustive and do not limit the present disclosure to the precise form

disclosed.

DETAILED DESCRIPTION

[0022] Embodiments of the systems and methods disclosed herein can provide for training a depth model to derive depth estimates from monocular images. For example, embodiments of the present disclosure may utilize sparse depth data from a range sensor as additional information to recover a scale factor and refine a self-supervised monocular depth estimation from monocular image sources.

[0023] Embodiments of the present disclosure utilize a combination of self-supervised photometric loss derived from images and sparse depth data from range sensors to supervise a scale of the images and align the scale with that of the range sensor. In various embodiments, the images are captured by monocular image sensors. As previously noted, perceiving aspects of the surrounding environment can represent different challenges depending on which sensors a device employs to support the endeavor. In particular, difficulties with using monocular image sensors to perceive depths in the surrounding environment can complicate the use of such sensors. For example, because a system trains and implements additional routines to derive the depth data from monocular images, difficulties can arise in relation to incomplete depth data and/or aberrations in the derived depth data from characteristics of the processing approach (e.g., scale ambiguity). The difficulties can cause the depth estimates to be unreliable for resolving aspects of the environment, thereby resulting in an incomplete situational awareness of the environment.

[0024] Therefore, embodiments herein employ a training architecture to support semi-supervised training from a combination of monocular images and sparse depth data to overcome the noted difficulties. Some embodiments disclosed herein include two stages that involve both self-supervised learning on images to obtain losses (e.g., photometric loss) and semi-supervised learning to recover the scale. Whereas various approaches to supervised training of a depth model may use comprehensive depth maps (e.g., nearly per-pixel annotations) that correlate with images or may use stereo images to provide self-supervision of a training process, the embodiments herein implement the training architecture on monocular images in the first stage, thereby avoiding a need for specialized sensors, and sparse depth data in the second stage as additional ground truth information thereby using depth labels from less expensive range sensors. Thus, the embodiments herein improve the training process by using standard monocular image sensors (e.g., cameras) to capture monocular images, and supplementing the self-supervised training with the second stage that refines the depth model using sparse depth data to supervise the second stage.

[0025] For example, because annotated training data can be expensive to produce due to the use of expensive sensors (e.g., **64** beam LiDAR devices), and/or manual labeling processes, and because monocular video alone under a self-supervised process may result in scale ambiguities in the understanding of the trained depth model, embodiments herein overcome the noted difficulties by using sparse depth data to supervise a refining stage (i.e., the second stage) without reliance on more comprehensive/dense depth data from depth sensors having a high fidelity.

[0026] The sparse depth data is, in one embodiment, radar data (e.g., a point cloud) from a radar device that produces a sparsely populated perception of the surrounding environment. As opposed to providing a LiDAR device with a multiplicity of scanning beams that provide depth data at nearly a per-pixel level with a corresponding image, the depth system uses radar to produce sparse depth data. Consequently, the training architecture relies on the self-supervised training using monocular images, but improves the self-supervised process with the additional refinement second stage that uses less complex and, thus, more easily acquired depth data to improve the training process. In this way, the semi-supervised training with weak supervision improves the ability of the depth model to infer metrically accurate depths without using extensively annotated training data.

[0027] However, because radar depth data can be noisy, instead of applying semi-supervised training on a pixel-by-pixel basis (e.g., pixel-wise) as with LiDAR depth data, embodiments disclosed herein estimate a single scale the radar depth data (e.g., a single scale for an entire point cloud) and refines the self-supervised depth model using the estimated scale by aligning a scale factor of the monocular image with the estimate scale. The determination may be based, for example, a measure of central

tendency of scale of points in of the point cloud. A measure of central tendency, as used herein, refers to a measure of a central or typical value for probability distribution and may refer to, for example, an arithmetic average (or mean), weighted average, median, mode, and the like. Estimating scale on a point-basis (e.g., a scale for each point cloud) may be provide for improved performance over pixel-wise application of radar depth data, since the estimate averages out potential sources of error (e.g., uncertainty due to noise in the detection results). In some embodiments, weights may be determined based on uncertainty in the radar depth data, for example, higher uncertainty corresponds to smaller weights. The weighted average is updated with new valid depth pixel, relative to the corresponding monocular depth estimate. In some embodiments, a single scale may be estimated for multiple sets of radar data (e.g., a plurality point clouds across varying environment scenes), for example, where the uncertainty or error in the radar data is greater than a determined threshold. The estimated scale for the plurality of sets (referred to herein as an aggregate scale) may train the depth model to enforce consistency in scale across self-supervised monocular outputs. Accordingly, depth maps generated by a learned depth model for different timesteps may be scale-consistent, if not scale-aware. Accordingly, embodiments herein provide for scale-aware self-supervised depth models by leveraging radar data to obtain a single scale factor to produce metrically accurate depth estimates in monocular images.

[0028] Accordingly, embodiments disclosed herein implement a two-stage training process with the first stage including self-supervised learning on images and a second stage including the semi-supervised learning that uses sparse depth data to generate a single scale factor to refine training of the depth model. In this way, the disclosed embodiments provide for semi-supervised training that improves the understanding of the depth model while avoiding a need for denser annotated depth data to provide scale-aware depth estimates.

[0029] The systems and methods disclosed herein may be implemented with any of a number of different vehicles and vehicle types. For example, the systems and methods disclosed herein may be used with automobiles, trucks, motorcycles, recreational vehicles and other like on- or off-road vehicles. In addition, the principals disclosed herein may also extend to other vehicle types as well.

[0030] An example hybrid electric vehicle (HEV) in which embodiments of the disclosed technology may be implemented is illustrated in FIG. 1. Although the example described with reference to FIG. 1 is a hybrid type of vehicle, the systems and methods for process of semi-supervised scale-aware learning of a depth model for monocular depth estimation can be implemented in other types of vehicle including gasoline- or diesel-powered vehicles, fuel-cell vehicles, electric vehicles, or other vehicles.

[0031] FIG. 1 illustrates a drive system of vehicle 10 that may include an internal combustion engine 14 and one or more electric motors 22 (which may also serve as generators) as sources of motive power. Driving force generated by the internal combustion engine 14 and motors 22 can be transmitted to one or more wheels 34 via a torque converter 16, a transmission 18, a differential gear device 28, and a pair of axles 30.

[0032] Vehicle 10 may be driven/powered with either or both of engine 14 and motor(s) 22 as the drive source for travel. For example, a first travel mode may be an engine-only travel mode that only uses internal combustion engine 14 as the source of motive power. A second travel mode may be an EV travel mode that only uses the motor(s) 22 as the source of motive power. A third travel mode may be a hybrid electric vehicle (HEV) travel mode that uses engine 14 and the motor(s) 22 as the sources of motive power. In the engine-only and HEV travel modes, vehicle 10 relies on the motive force generated at least by internal combustion engine 14, and clutch 15 may be included to engage engine 14. In the EV travel mode, vehicle 10 is powered by the motive force generated by motor 22 while engine 14 may be stopped and clutch 15 disengaged.

[0033] Engine 14 can be an internal combustion engine such as a gasoline, diesel or similarly powered engine in which fuel is injected into and combusted in a combustion chamber. A cooling system 12 can be provided to cool the engine 14 such as, for example, by removing excess heat from engine 14. For example, cooling system 12 can be implemented to include a radiator, a water pump and a series of cooling channels. In operation, the water pump circulates coolant through the engine 14 to absorb

excess heat from the engine. The heated coolant is circulated through the radiator to remove heat from the coolant, and the cold coolant can then be recirculated through the engine. A fan may also be included to increase the cooling capacity of the radiator. The water pump, and in some instances the fan, may operate via a direct or indirect coupling to the driveshaft of engine **14**. In other applications, either or both the water pump and the fan may be operated by electric current such as from battery **44**. [0034] An output control circuit **14A** may be provided to control drive (output torque) of engine **14**. Output control circuit **14A** may include a throttle actuator to control an electronic throttle valve that controls fuel injection, an ignition device that controls ignition timing, and the like. Output control circuit **14A** may execute output control of engine **14** according to a command control signal(s) supplied from electronic control unit **50**, described below. Such output control can include, for example, throttle control, fuel injection control, and ignition timing control.

[0035] Motor **22** can also be used to provide motive power in vehicle **10** and is powered electrically via battery **44**. Battery **44** may be implemented as one or more batteries or other power storage devices including, for example, lead-acid batteries, lithium ion batteries, capacitive storage devices, and so on. Battery **44** may be charged by a battery charger **45** that receives energy from internal combustion engine **14**. For example, an alternator or generator may be coupled directly or indirectly to a drive shaft of internal combustion engine **14** to generate an electrical current as a result of the operation of internal combustion engine **14**. A clutch can be included to engage/disengage the battery charger **45**. Battery **44** may also be charged by motor **22** such as, for example, by regenerative braking or by coasting during which time motor **22** operate as generator.

[0036] Motor **22** can be powered by battery **44** to generate a motive force to move vehicle **10** and adjust vehicle speed. Motor **22** can also function as a generator to generate electrical power such as, for example, when coasting or braking. Battery **44** may also be used to power other electrical or electronic systems in the vehicle. Motor **22** may be connected to battery **44** via an inverter **42**. Battery **44** can include, for example, one or more batteries, capacitive storage units, or other storage reservoirs suitable for storing electrical energy that can be used to power motor **22**. When battery **44** is implemented using one or more batteries, the batteries can include, for example, nickel metal hydride batteries, lithium ion batteries, lead acid batteries, nickel cadmium batteries, lithium ion polymer batteries, and other types of batteries.

[0037] An electronic control unit **50** (described below) may be included and may control the electric drive components of the vehicle as well as other vehicle components. For example, electronic control unit **50** may control inverter **42**, adjust driving current supplied to motor **22**, and adjust the current received from motor **22** during regenerative coasting and braking. As a more particular example, output torque of the motor **22** can be increased or decreased by electronic control unit **50** through inverter **42**.

[0038] A torque converter **16** can be included to control the application of power from engine **14** and motor **22** to transmission **18**. Torque converter **16** can include a viscous fluid coupling that transfers rotational power from the motive power source to the driveshaft via the transmission. Torque converter **16** can include a conventional torque converter or a lockup torque converter. In other embodiments, a mechanical clutch can be used in place of torque converter **16**.

[0039] Clutch **15** can be included to engage and disengage engine **14** from the drivetrain of vehicle **10**. In the illustrated example, a crankshaft **32**, which is an output member of engine **14**, may be selectively coupled to the motor **22** and torque converter **16** via clutch **15**. Clutch **15** can be implemented as, for example, a multiple disc type hydraulic frictional engagement device whose engagement is controlled by an actuator such as a hydraulic actuator. Clutch **15** may be controlled such that its engagement state is complete engagement, slip engagement, and complete disengagement complete disengagement, depending on the pressure applied to the clutch. For example, a torque capacity of clutch **15** may be controlled according to the hydraulic pressure supplied from a hydraulic control circuit (not illustrated). When clutch **15** is engaged, power transmission is provided in the power transmission path between crankshaft **32** and torque converter **16**. On the other hand, when clutch **15** is disengaged, motive power from engine **14** is not delivered to the torque converter **16**. In a

slip engagement state, clutch **15** is engaged, and motive power is provided to torque converter **16** according to a torque capacity (transmission torque) of the clutch **15**.

[0040] As alluded to above, vehicle **10** may include electronic control unit **50**. Electronic control unit **50** may include circuitry to control various aspects of the vehicle operation. Electronic control unit **50** may include, for example, a microcomputer that includes a one or more processing units (e.g., microprocessors), memory storage (e.g., RAM, ROM, etc.), and I/O devices. The processing units of electronic control unit **50**, execute instructions stored in memory to control one or more electrical systems or subsystems in the vehicle. Electronic control unit **50** can include a plurality of electronic control units such as, for example, an electronic engine control module, a powertrain control module, a transmission control module, a suspension control module, a body control module, and so on. As a further example, electronic control units can be included to control systems and functions such as doors and door locking, lighting, human-machine interfaces, cruise control, telematics, braking systems (e.g., ABS or ESC), battery management systems, and so on. These various control units can be implemented using two or more separate electronic control units or using a single electronic control unit.

[0041] In the example illustrated in FIG. **1**, electronic control unit **50** receives information from a plurality of sensors included in vehicle **10**. For example, electronic control unit **50** may receive signals that indicate vehicle operating conditions or characteristics, or signals that can be used to derive vehicle operating conditions or characteristics. These may include, but are not limited to accelerator operation amount (ACC), a revolution speed (NE), of internal combustion engine **14** (engine RPM), a rotational speed of the motor **22** (motor rotational speed), and vehicle speed, NV. These may also include torque converter **16** output (e.g., output amps indicative of motor output), brake operation amount/pressure, B, battery (i.e., the charged amount for battery **44** detected by a system on chip (SOC) sensor). Accordingly, vehicle **10** can include a plurality of sensors **52** that can be used to detect various conditions internal or external to the vehicle and provide sensed conditions to electronic control unit **50** (which, again, may be implemented as one or more individual control circuits). In one embodiment, sensors **52** may be included to detect one or more conditions directly or indirectly such as, for example, fuel efficiency (E.sub.F), motor efficiency (E.sub.MG), hybrid (e.g., ICE **14** and MG **12**) efficiency, acceleration, ACC, etc.

[0042] Additionally, one or more sensors **52** can be configured to detect, and/or sense position and orientation changes of the vehicle **10**, such as, for example, based on inertial acceleration, trajectory, and so on. In one or more arrangements, electronic control unit **50** can obtain signals from vehicle sensor(s) including accelerometers, one or more gyroscopes, an inertial measurement unit (IMU), a dead-reckoning system, a global navigation satellite system (GNSS), a global positioning system (GPS), a navigation system, and/or other suitable sensors. In one or more arrangements, electronic control unit **50** receives signals from a speedometer to determine a current speed of the vehicle **10**.

[0043] Sensors **52** may be included to detect not only vehicle conditions but also to detect environment conditions external and/or internal to the vehicle. Sensors that might be used to detect external conditions can include, for example, distance measuring sensors or range sensors (e.g., sonar, radar, LiDAR, infra-red cameras, and the like), vehicle proximity sensors, and image sensors (e.g., cameras or other image sensors). In some embodiments, cameras can be high dynamic range (HDR) cameras or infrared (IR) cameras. Image and range sensors can be used to detect the environment surrounding the vehicle **10**, for example, traffic signs, road curvature, obstacles, and so on. Still other sensors may include those that can detect road grade.

[0044] In some embodiments, one or more of the sensors **52** may include their own processing capability to compute the results for additional information that can be provided to electronic control unit **50**. In other embodiments, one or more sensors may be data-gathering-only sensors that provide only raw data to electronic control unit **50**. In further embodiments, hybrid sensors may be included that provide a combination of raw data and processed data to electronic control unit **50**. Sensors **52** may provide an analog output or a digital output. Additionally, as alluded to above, the one or more sensors **52** can be configured to detect, and/or sense in real-time. As used herein, the term “real-time”

means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep up with some external process.

[0045] According to an embodiment, vehicle **10** can be an autonomous vehicle. As used herein, “autonomous vehicle” can refer to a vehicle that is configured to operate in an autonomous operational mode. “Autonomous operational mode” can refer to the use of one or more computing systems of the vehicle **10** to navigate and/or maneuver vehicle **10** along a travel route with a level of input from a human driver which can vary with the operational mode, for example, based on information detected by sensors **52**. As such, vehicle **10** can have a plurality of autonomous operational modes. In some embodiments, vehicle **10** can have an unmonitored autonomous operational mode, meaning that one or more computing systems are used to maneuver vehicle **10** along a travel route fully autonomously, requiring no input or supervision required from a human driver.

[0046] Alternatively, or in addition to the above-described modes, vehicle **10** can have one or more semi-autonomous operational modes. “Semi-autonomous operational mode” can refer to mode whereby a portion of the navigation and/or maneuvering of vehicle **10** along a travel route is performed by one or more computing systems, for example, based on information detected by sensors **52**, while a portion of navigation and/or maneuvering of vehicle **10** along the travel route is performed by a human driver. One example of a semi-autonomous operational mode is an adaptive cruise control system. In such case, the speed of vehicle **10** can be automatically adjusted to maintain a safe distance from a vehicle ahead based on data received from on-board sensors, but vehicle **10** is otherwise operated manually by a human driver. Another example of a semi-autonomous operational mode include Advanced Driver-Assistance Systems (ADAS), such as, forward/rear collision detection and warning systems, pedestrian detection systems, etc.

[0047] The example of FIG. **1** is provided for illustration purposes only as an example of vehicle systems with which embodiments of the disclosed technology may be implemented. Embodiments herein are not limited to automobiles. For example, embodiments herein may be implemented in any electronic/robotic device or another form of powered transport that, for example, perceives an environment according to environment sensors. Additionally, embodiments herein may be implemented in a statically mounted device, an embedded device, or another device that uses environment sensor data to derive depth information about a scene or that separately trains the depth model for deployment in such a device. For example, embodiments herein may be implemented in a server (e.g., a physical, dedicated sever or a cloud-based server coupled to a database resident on network), and the resulting depth model may be communicated to other remote devices for use in autonomous and/or semi-autonomous operational modes. Thus, one of ordinary skill in the art reading this description will understand how the disclosed embodiments can be implemented with any vehicle, robotic, and/or computation platform.

[0048] FIG. **2** illustrates an example architecture for semi-supervised scale-aware learning in accordance with one embodiment of the systems and methods described herein. In this example, system **200** includes a semi-supervised learning circuit **210**, the plurality of sensors **52**, and one or more vehicle systems **220**. Sensors **52** and vehicle systems **220** can communicate with semi-supervised learning circuit **210** via a wired or wireless communication interface. semi-supervised learning circuit **210** can be implemented, for example, as an ECU or as part of an ECU such as, for example ECU **50** of vehicle. In other embodiments, semi-supervised learning circuit **210** can be implemented independently of an ECU, for example, as a standalone computing component (e.g., cloud-based server resident on a network, physical or dedicated server, software-as-a-service (Saas), distributed computing service, etc.).

[0049] Semi-supervised learning circuit **210**, in this example, includes a communication circuit **201**, a decision circuit **203** (including a processor **206** and memory **208** in this example), data store **205**, and a power supply **212**. Components of semi-supervised learning circuit **210** are illustrated as communicating with each other via a data bus, although other communication in interfaces can be included.

[0050] Processor **206** can include a GPU, CPU, microprocessor, or any other suitable processing system. Memory **208** may include one or more various forms of memory or data storage (e.g., flash, RAM, etc.) that may be used to store the calibration parameters, images (analysis or historic), point parameters, instructions and variables for processor **206** as well as any other suitable information. Memory **208** can be made up of one or more modules of one or more different types of memory and may be configured to store data and other information as well as operational instructions that may be used by the processor **206** to control semi-supervised learning circuit **210**. For example, memory **208** may store one or more modules, for example, as computer-readable instructions that when executed by the processor **206** cause the processor **206** to perform the various functions and operations of the semi-supervised learning circuit **210**.

[0051] Data store **205** may be an electronic data structure such as a database. The data store **205** may be stored in the memory **208** or another memory. The data store **205** may be configured with routines that can be executed by the processor **206** for analyzing stored data, providing stored data, organizing stored data, and so on. Accordingly, the data store **205** may store data used by the processor **206** in executing various functions. The data store **205** may also receive and store data from sensors **52** and/or vehicle systems **220**, for example, for use by the processor **206** in executing operations and functions of the semi-supervised learning circuit **210**.

[0052] Although the example of FIG. **2** is illustrated using processor and memory circuitry, as described below with reference to circuits disclosed herein, decision circuit **203** can be implemented utilizing any form of circuitry including, for example, hardware, software, or a combination thereof. By way of further example, one or more processors, controllers, ASICs, PLAS, PALs, CPLDs, FPGAs, logical components, software routines or other mechanisms might be implemented to make up a semi-supervised learning circuit **210**.

[0053] Communication circuit **201** either or both a wireless transceiver circuit **202** with an associated antenna **214** and a wired I/O interface **204** with an associated hardwired data port (not illustrated). As this example illustrates, communications with semi-supervised learning circuit **210** can include either or both wired and wireless communications circuits **201**. Wireless transceiver circuit **202** can include a transmitter and a receiver (not shown) to allow wireless communications via any of a number of communication protocols such as, for example, WiFi, Bluetooth, near field communications (NFC), Zigbee, and any of a number of other wireless communication protocols whether standardized, proprietary, open, point-to-point, networked or otherwise. Antenna **214** is coupled to wireless transceiver circuit **202** and is used by wireless transceiver circuit **202** to transmit radio signals wirelessly to wireless equipment with which it is connected and to receive radio signals as well. These RF signals can include information of almost any sort that is sent or received by semi-supervised learning circuit **210** to/from other entities, such as, sensors **52**, vehicle systems **220**, edge devices (e.g., remote network node, edge computing devices, etc.), and the cloud (e.g., cloud-based server resident on network, etc.).

[0054] Wired I/O interface **204** can include a transmitter and a receiver (not shown) for hardwired communications with other devices. For example, wired I/O interface **204** can provide a hardwired interface to other components, including sensors **52** and vehicle systems **220**. Wired I/O interface **204** can communicate with other devices using Ethernet or any of a number of other wired communication protocols whether standardized, proprietary, open, point-to-point, networked or otherwise.

[0055] Power supply **210** can include one or more of a battery or batteries (such as, e.g., Li-ion, Li-Polymer, NiMH, NiCd, NiZn, and NiH.sub.2, to name a few, whether rechargeable or primary batteries), a power connector (e.g., to connect to vehicle supplied power, etc.), an energy harvester (e.g., solar cells, piezoelectric system, etc.), or it can include any other suitable power supply.

[0056] Sensors **52** can include, for example, those described above with reference to the example of FIG. **1**. Sensors **52** can include additional sensors that may or not otherwise be included on a standard vehicle **10** with which the semi-supervised scale-aware learning system **200** is implemented. In the illustrated example, sensors **52** include one or more vehicle sensors **52A**. The vehicle sensor(s) **52A** can detect, determine, and/or sense information about the vehicle **10**. For example, vehicle sensor(s)

52A can be configured to detect, and/or sense position and orientation changes of the vehicle **10**, such as, for example, based on inertial acceleration. In various embodiments, the vehicle sensor(s) **52A** can include one or more accelerometers (e.g., to detect roll, pitch, and yaw of the vehicle), one or more gyroscopes, an inertial measurement unit (IMU). The vehicle sensor(s) **52A** can also be configured to detect, and/or sense one or more characteristics of the vehicle **10**. For example, vehicle sensor(s) **52A** may include vehicle acceleration sensors, vehicle speed sensors, throttle and brake pedal sensors, steering sensors, wheelspin sensors (e.g., one for each wheel), tire pressure monitoring system (TPMS), proximity sensors, etc.

[0057] Sensors **52** also includes environment sensors **52B** configured to acquire, and/or sense environment data surrounding the vehicle and/or internal to the vehicle. Environment data includes data or information about the external and/or internal environment in which vehicle **10** is located or one or more portions thereof. For example, the one or more environment sensors **52B** can be configured to detect, quantify and/or sense obstacles in at least a portion of the external environment of the vehicle **10** and/or information/data about such obstacles. Such obstacles may be stationary objects and/or dynamic objects. The one or more environment sensors **52B** can be configured to detect, measure, quantify and/or sense other things in the external environment, such as, for example, other vehicles, pedestrians, lane markers, signs, traffic lights, traffic signs, lane lines, crosswalks, curbs proximate the vehicle **10**, off-road objects, etc. As another example, alone or in addition to the above, the one or more environment sensors **52B** can be configured to detect, quantify and/or sense obstacles in at least a portion of the internal environment of the vehicle **10** and/or information/data about such environment.

[0058] Environment sensors **52B** includes one or more range sensors such as, but not limited to, LiDAR **240**, radar **242**, sonar **244**, IR sensors and other like sensors that can be used to gather data (e.g., data **250**) to measure distances and closing rates to various external objects. Environment sensors **52B** also includes one or more image sensors **246**, such as cameras (e.g., visible light cameras, IR cameras, thermal cameras, ultrasound cameras, and other cameras) or other image sensors, to capture video images **248** of the environment around the vehicle. Information from image sensors **246** can be used, for example, by vehicle systems **220** to determine information about the environment surrounding the vehicle **10** including, for example, information regarding other objects surrounding vehicle **10**. For example, image sensors **246** capture video images **248** from which vehicle systems **220** may be able to recognize landmarks or other features (including, e.g., street signs, traffic lights, etc.), slope of the road, lines on the road, curbs, objects to be avoided (e.g., other vehicles, pedestrians, bicyclists, etc.) and other landmarks or features. Information from image sensors **246** can be used in conjunction with other information such as map data or information from navigation system **222** to determine, refine or verify vehicle location.

[0059] In various embodiments, each image sensor **246** is a monocular image sensor configured to capture monocular videos as a plurality of frames in the form of monocular images (e.g., images **248**). As described herein, a “monocular image” is an image from a single (e.g., monocular) camera, and encompasses a field-of-view (FOV) of at least a portion of the surrounding environment (e.g., a subregion of the surrounding environment). A monocular image may not include any explicit additional modality indicating depth nor any explicit corresponding image from another camera from which the depth can be derived (e.g., no stereo image sensor pair). In contrast to a stereo image, that may integrate left and right images from separate cameras mounted side-by-side to provide an additional depth channel, a monocular image does not include explicit depth information such as disparity maps derived from comparing the stereo images pixel-by-pixel. Instead, a monocular image may implicitly provide depth information in the relationships of perspective and size of elements depicted therein. The monocular image may be of a forward-facing (e.g., the direction of travel), 60-degree FOV, 90-degree FOV, 120-degree FOV, a rear/side facing FOV, or some other subregion based on the positioning and characteristics of the image sensor **246** on a vehicle (e.g. vehicle **10**).

[0060] Additional sensors **52C** may be provided as well. Various sensors **52** may be used to provide input to semi-supervised learning circuit **210** and other vehicle systems **220** so that the systems have

information useful to operate in an autonomous, semi-autonomous or manual mode.

[0061] Vehicle systems **220** can include any of a number of different vehicle components or subsystems used to control or monitor various aspects of a vehicle, e.g., vehicle **10**, and its performance. In this example, vehicle systems **220** include a GPS or other vehicle positioning system **222**; throttle/brake circuits **224** (e.g., to control operation of motor/generator **22**, to control the operation of the engine including internal combustion engine **14**, and/or control operation of a brake system to decelerate the vehicle); ADAS systems **228** (e.g., to control operation of obstacle detection and warning, forward and/or rear collision detection, etc.); autonomous and/or semi-autonomous control systems **226** (e.g., to control operation of navigation and/or maneuvering according to autonomous or semi-autonomous operational modes), and other vehicle systems **282**.

[0062] During operation, semi-supervised learning circuit **210** can receive information from various vehicle sensors and/or vehicle systems **220** to perform semi-supervised scale-aware learning of a depth model for depth estimation. Communication circuit **201** can be used to transmit and receive information between semi-supervised learning circuit **210** and sensors **52**, and semi-supervised learning circuit **210** and vehicle systems **220**. Also, sensors **52** may communicate with vehicle systems **220** directly or indirectly (e.g., via communication circuit **201** or otherwise).

[0063] In various embodiments, communication circuit **201** can be configured to receive data and other information from sensors **52** that is used in determining training and deploying a scale aware depth model according to embodiment disclosed herein. Additionally, communication circuit **201** can be used to send data and information to various vehicle systems **220** as part of performing operations of the respective vehicle systems **220**. The decision regarding what action to take via these various vehicle systems **220** can be made based on the information detected by sensors **52** and application of a trained scale-aware depth model. Examples of this are described in more detail below.

[0064] Once trained, the scale-aware depth model may be employed, for example, to analyze images **248** (e.g., monocular images) from an image sensor **246** and provide a depth map **308** to additional vehicle systems **220**. For example, processor **206** may generate a scale-aware depth map from images **248** and provide the depth map to the vehicle systems **220**. For example, a scale-aware depth map may be provided to ADAS system **228** (e.g., to control operation of obstacle detection and warning, forward and/or rear collision detection, etc.), autonomous and/or semi-autonomous control systems **226** (e.g., to control operation of navigation and/or maneuvering according to autonomous or semi-autonomous operational modes), and other operation of the vehicle systems **220**.

[0065] In another example, a scale-aware depth map may be communicated (e.g., via communication circuit **301**) to a remote system (e.g., cloud-based systems, edge devices, etc.) as, for example, a mechanism for mapping the surrounding environment or for other purposes (e.g., traffic reporting, etc.). As yet another example, the depth map **308** may be communicated (e.g., via communication circuit **301**) to map locations of obstacles in the surrounding environment and plan a trajectory that safely navigates the obstacles. Thus, the depth map may be used to control the vehicle **10** to navigate through the surrounding environment.

[0066] FIG. 3 illustrates a schematic block diagram of an example semi-supervised learning circuit **210** in accordance with embodiments of the systems and methods described herein. As alluded to above, semi-supervised learning circuit **210** may be provided to train and implement a scale-aware depth model to process images (e.g., images **248**) and provide depth estimates for an environment (e.g., objects, surfaces, etc.) depicted therein.

[0067] Memory **208** may be configured to store, for example, a network module **320** and a training module **330**. The modules **320** and **330** are, for example, computer-readable instructions that when executed by the processor **206** cause the processor **206** to perform the various functions disclosed herein. The data store **205** may be configured to store data used by the modules **220** and **230** in executing various functions. In the illustrative example of FIG. 3, the data store **205** stores training data **302**, a depth model **304**, a depth map(s) **308**, a pose model **306**, a transformation(s) **310**, and depth data **312** along with, for example, other information that is used by the modules **320** and **330**.

[0068] The training data **302** generally includes one or more monocular videos that are comprised of a

plurality of images **248**. The monocular images **248** may implicitly provide depth information (e.g., depth values) in the relationships of perspective and size of elements depicted therein from which the depth model **304** may derive the depth map **308**. The depth map **308** may comprise a plurality of pixels arranged in two dimensional space (e.g., X and Y) and comprising a depth value derived from the monocular image **248**.

[0069] The monocular video may also include observations of many different scenes. For example, as the image sensor **246** progresses through an environment, perspectives of objects and features in the environment change, and the depicted objects/features themselves also change, thereby depicting separate scenes (e.g., particular combinations of objects/features). Thus, the semi-supervised learning circuit **210** may extract particular training pairs of monocular images from the monocular video for training. For example, the semi-supervised learning circuit **210** generates pairs from the video so that the pairs of images are of the same or similar scene. As used herein, a “similar scene” is a scene having a threshold amount of overlap in features/objects depicted therein as compared to a scene of another image. As should be appreciated, the video includes a series of monocular images **248** that are taken in succession according to a configuration of the image sensor **246**. Thus, the image sensor **246** may generate the images **248** of the video at regular intervals, such as 10 Hz. That is, the image sensor **246** generates an image **248** at a frequency, which may be, for example, 10 Hz. While a specific frequency is describes, any frequency may be used, for example, 30 Hz, 60 Hz, etc. While higher frequencies may require faster algorithms, since the interval between updates is smaller, there are no hard limits or requirements on the frequency at which images **248** are generated.

[0070] The semi-supervised learning circuit **210** need not generate pairs from successive images (e.g., adjacent frames). The semi-supervised learning circuit **210** may pair separate images of the same or similar scene that are not successive as training images. Thus, in one example, the semi-supervised learning circuit **210** may pair every other image depending on the capture rate. In another example, the semi-supervised learning circuit **210** may pair every fifth image as a training pair. The greater the timing difference in the video between the pairs, the more pronounced a difference in image sensor position; however, this may also result in fewer shared (e.g., overlapped) features/objects between the images. As such, as previously noted, the pairs of training images are of a same scene and are generally constrained, in one or more embodiments, to be within a defined number of frames (e.g., 5 or fewer) to ensure correspondence of an observed scene between the monocular training images. In any case, pairs of training images have the attributes of being monocular images from a monocular video that are separated by some interval of time (e.g., 0.1 s) such that a perspective of the image sensor changes between the pair of training images as a result of motion of the image sensor through the environment while generating the video.

[0071] Additionally, the training data **302**, in one or more embodiments, further includes depth data **312** received from a range sensor (e.g., radar sensor **246**). The depth data **312** includes depth measures indicating distances from an environment sensor to features in the surrounding environment. The depth data **312** may be provided as a point cloud, which is sparse or generally incomplete for a corresponding scene such that only sparsely distributed points within a scene are annotated by the depth measures as opposed to a depth map that generally provides comprehensive depths for each separate depicted pixel. That is, only those points for which a depth measure is detected by the range sensor are included in point cloud. The point cloud is generally provided in a 3D coordinate space (e.g., X, Y, and Z). Additionally, the depth data may be noisy such that those annotated points within the scene include a degree of error or uncertainty in the depth measurements.

[0072] While the depth data **312** is generally described as originating from a radar sensor, the embodiments herein are applicable to depth data from LiDAR or another range sensors. Furthermore, the depth data **312** itself generally includes depth/distance information relative to a point of origin such as the range sensor, and may also include coordinates (e.g., x, y within an image) corresponding with separate depth measurements. An unscaled point cloud may be accurate up to a scale value, and when the point cloud is multiplied by its corresponding scale value it produces metric estimates. This scale is a conversion unit that transforms from the unknown monocular scale to metric scale.

[0073] For example, a depth map derived from an image of a scene captured by an image sensor includes a plurality of annotated points generally corresponding to an associated monocular image on a per pixel basis. Thus, the depth map includes separate annotated points for each pixel. By comparison, a range sensor (e.g., a radar sensor **242**) may generate a 3D point cloud that can be projected onto a 2D image plane. The projection of the 3D point cloud onto the image plane provides a sparse number of pixels arranged on the 2D plane having visual attributes based on the depth measurements. For example, a radar sensor **242** may generate a point cloud that when projected to the image plane provides 100 points that are annotated with depth measurements. Radar and LiDAR sensors generate sparse point clouds, with point clouds generated by LiDAR sensors being less sparse than those from radar sensors. The point clouds are sparse because they are obtained by projecting 3D points onto an image plane. Some image pixels won't have any projected point, so they don't have any valid information. Because radar sensors produce fewer points than LiDAR, there will be less information to project and, by extension, fewer valid depth pixels. Furthermore, in the case of radar sensors, the depth measurements for each point is noisy, inherently injecting uncertainty to each depth measure. For example, the noise values can be due to range errors (e.g., a point is closer/further away than measured), or radial errors (e.g., the point projection is not where it should be). Radar sensors are much noisier radially, relative to LiDAR sensors. Even though the point cloud for a range sensor includes sparse and noisy depth measurements, the information provides for sufficient supervision to facilitate overcoming scale ambiguities within the depth model **304** when used as an additional refinement process for training in combination with a self-supervision process.

[0074] As an additional comparison, the depth data within the image, based on depth model **304**, is sufficiently dense to convey details of existing features/objects such as vehicles, etc. However, within the projection of the point cloud from the range sensor onto the image plane, the depth data is sparse, noisy and imprecise. The depth data vaguely characterizes the corresponding scene according to distributed points across the scene that do not generally provide detail of specific features/objects depicted therein and includes inherent uncertainty in the depth measurement. Thus, this sparse depth data that is noisy across the scene may not provide enough data for some purposes such as object classification but does provide sufficient information to determine distances of an object and supervise a refinement stage of training the depth model **304**.

[0075] Furthermore, radar sensors offer cost advantage over other sources of depth information, such as velocity measurements, inertial sensors, LiDAR, camera extrinsic, etc. For example, in the case of LiDAR sensors, while LiDAR may offer more precise depth measurements, the depth measurements collected by LiDAR may represent a significant cost to acquire over a monocular video on an image-by-image basis. These costs and other difficulties generally relate to an expense of a LiDAR sensor that include producing a denser point cloud (e.g., 64 separate beams), difficulties in calibrating LiDAR devices with the monocular camera, storing large quantities of data associated with such a point cloud for each separate image, and so on. Furthermore, radar devices are generally cheaper and consume less energy. Additionally, radar devices are smaller in size for easier placement on or within a vehicle.

[0076] Returning to FIG. **3**, the data store **205** stores the depth model **304** for producing the depth map **308** and the pose model **306** for producing the transformation **310**. Both of the models **304** and **306** are, in various embodiments, machine learning algorithms. However, the particular form of either model is generally distinct. That is, for example, the depth model **304** is a machine learning algorithm that accepts an electronic input in the form of a single monocular image and produces the depth map **308** as a result of processing the monocular image. The exact form of the depth model **304** may vary according to implementation but may be conveyed as a convolutional type of neural network.

[0077] FIG. **4** illustrates a schematic block diagram of an example depth model **400** according to embodiments disclosed herein. The depth model **400** is an example implementation of depth model **304** of FIG. **3**.

[0078] In the example illustrated in FIG. **4**, the depth model **400** includes an encoder/decoder architecture. The encoder/decoder architecture may include a set of neural network layers including convolutional components **410** (e.g., 2D and/or 3D convolutional layers forming an encoder) that flow

into convolutional components **420** (e.g., 2D and/or 3D deconvolutional layers forming a decoder). In one approach, the encoder accepts an image **401** (e.g., a monocular image **248**, for example, from the training data **302**), as an electronic input and processes the image to extract features therefrom. The features are, in general, aspects of the image that are indicative of spatial information that the image intrinsically encodes. As such, encoding layers that form the encoder function to, for example, fold (i.e., adapt dimensions of the feature map to retain the features) encoded features into separate channels, iteratively reducing spatial dimensions of the image while packing additional channels with information about embedded states of the features. Thus, the addition of the extra channels avoids the lossy nature of the encoding process and facilitates the preservation of more information (e.g., feature details) about the original monocular image.

[0079] As illustrated in FIG. **4**, the encoder **410** comprises multiple encoding layers formed from a combination of two-dimensional (2D) convolutional layers, packing blocks, and residual blocks. Moreover, the separate encoding layers generate outputs in the form of encoded feature maps (also referred to as tensors), which the encoding layers provide to subsequent layers in the depth model **400**. As such, the encoder includes a variety of separate layers that operate on the monocular image, and subsequently on derived/intermediate feature maps that convert the visual information of the monocular image into embedded state information in the form of encoded features of different channels.

[0080] The decoder **420** may unfold (e.g., adapt dimensions of the tensor to extract the features) the previously encoded spatial information in order to derive a depth map (e.g., depth map **308**) according to learned correlations associated with the encoded features. For example, the decoding layers may function to up-sample, through sub-pixel convolutions and other mechanisms, the previously encoded features into the depth map, which may be provided at different resolutions **430**. In some embodiments, the decoding layers comprise unpacking blocks, two-dimensional convolutional layers, and inverse depth layers that function as output layers for different scales of the feature/depth map. The depth map may be a data structure corresponding to the input image that indicates distances/depths to objects/features represented therein. Additionally, in various embodiments, the depth map may be a tensor with separate data values indicating depths for corresponding locations in the image on a per-pixel basis.

[0081] The depth model **400** may further include skip connections **440** for providing residual information between the encoder and the decoder to facilitate memory of higher-level features between the separate components. While a particular encoder/decoder architecture is discussed, as previously noted, the depth model **304**, in various approaches, may take different forms and generally functions to process the monocular images and provide depth maps that are per-pixel estimates about distances of objects/features depicted in the images.

[0082] FIG. **5** illustrates a schematic block diagram of an example pose model **500** according to embodiments disclosed herein. The pose model **500** is an example implementation of the pose model **306** of FIG. **3**.

[0083] In the example of FIG. **5**, the pose model **500** accepts two monocular images **501** (e.g., a training pair), for example, from the training data **302** of the same or similar scene as the electronic input (e.g., image processed in the depth model) and processes the monocular images (I.sub.t, I.sub.s) to produce estimates of image sensor motion in the form of a set of 6 degree-of-freedom (DOF) transformations between the two images. The pose model **500** itself is, for example, a convolutional neural network (CNN) or another learning model that is differentiable and performs dimensional reduction of the input images to produce transformation **510** (e.g., an example of transformation **310**). In an example, the pose model **500** may include 7 stride-2 convolutions, a 1×1 convolution with 6*(N-1) output channels corresponding to 3 Euler angles and a 3-D translation for one of the images (source image I.sub.s), and global average pooling to aggregate predictions at all spatial locations. The transformation **510** may be, in some embodiments, a 6 DOF rigid-body transformation belonging to the special Euclidean group SE(3) that represents the change in pose between the pair of images provided as inputs to the model **500**. In any case, the pose model **500** performs a dimensional

reduction of the monocular images to derive the transformation 310 therefrom.

[0084] While the depth models 304 and the pose model 306 are shown as discrete units separate from the network module 320, the depth model 304 and the pose model 306 may be integrated with the network module 320. That is, the network module 320 may function to execute various processes of the models 304 and/or 306 and use various data structures of the models 304 and/or 306 in support of such execution. Accordingly, in one embodiment, the network module 320 includes instructions that function to control the processor 206 to generate the depth map 308 using the depth model 304 and generate the transformation 310 using the pose model 306 as disclosed.

[0085] FIG. 6 is a diagram illustrating an example architecture for semi-supervised scale-aware learning of a depth model for monocular depth estimation. FIG. 6 illustrates an example training architecture formed in relation to the depth model 304 and the pose model 306, as previously described in connection with FIG. 3. FIG. 5 illustrates an example of a training architecture 600 that denotes various relationships between the depth model 304, the pose model 306, and inputs/outputs thereof. The architecture 600 may be implemented using, for example, one or more processors and memory elements such as, for example, processor 206 and memory 208.

[0086] As shown in FIG. 6, training data includes a training pair, for example, a first image (I.sub.t) 610 and a second image (I.sub.s) 620. The training pair may be received, for example, from the training data 302. The training data also includes depth data (D.sub.t) 312, for example, from the training data 302.

[0087] The training architecture 600 may be executed, for example, by the training module 330, which may include instructions that function to control the processor 206 to execute various actions associated with training the depth model 304 and the pose model 306. For example, the training module 330 may control the training according to a two-stage process, as shown in FIG. 6. The first stage may include a self-supervised training process that involves synthesizing an additional image using the transformation 310 and the depth map 308 produced from the models 304 and 306 operating on the images 610 and 620 in order to generate a self-supervised loss in the form of photometric loss 650. The second stage may include the same self-supervised training as the first stage supplemented with the depth data 312 to produce a supervised loss 660 in addition to the photometric loss 650. Thus, in the first stage and the second stage, the training module 330 may cause the network module 320 to execute the depth model 304 and the pose model 306 on the training data 250, as well as supplement the second stage with the additional supervised loss term 660.

[0088] The network module 320 may include instructions that function to control the processor 206 to execute various actions associated with the depth model 304 and the pose model 306. For example, in the first stage, the network module 320 may function to process the first image 610 of a pair of training images according to the depth model 304 to produce the depth map 308 for training. The network module 320, may also use the depth model 304 to generate the depth map 308 for additional purposes, once trained, such as resolving aspects of an environment for hazard avoidance, path planning, and so on (e.g., vehicle systems 220).

[0089] The network module 320 may also function to execute the pose model 306 to produce the transformation 310 (not illustrated in FIG. 6), which functions as a basis for synthesizing image (I.sub.t) 640 from which the training module 330 generates the photometric loss 650 as will be explained further below. Moreover, the network module 320 may execute the models 304 and 306 in concert (e.g., in parallel) for purposes of training, and at the direction of the training module 330. Thus, in addition to executing the model 304, the network module 320 may process the first image 610 and the second image 620 to generate the transformation 310 using the pose model 306. Thus, the network module 320 may operate to execute the models 304 and 306 over the training data 302, while the training module 330 functions to perform the explicit training processes such as generating the loss values and updating the models 304 and 306.

[0090] Once the network module 320 executes the models 304 and 306 over the images 610 and 620 to produce the depth map 308 and the transformation 310, the training module 330 generates the synthesized image (I.sub.t) 640. In various embodiments, the synthesized image 640 is, for example, a

synthesized version of the second image **620** according to the depth map **308** and the transformation **310**. That is, the process of semi-supervised training of the depth model **308** in the structure-from-motion (SfM) context in the first and second stages involves synthesizing a different image from the first image **610** that is of the same or similar scene from a different pose and/or timestamp of the image sensor. The training module **330** generates the synthesized image **640** using a machine learning algorithm such as a generative neural network (e.g., encoder/decoder architecture, a generative adversarial network (GAN), an autoencoder, etc.), a convolutional neural network (CNN), or another suitable architecture that accepts the depth map **308** and the transformation **310** as input and produces the synthesized image **640** as output.

[0091] From the synthesized image **640**, the training module **330** can generate the photometric loss **650** and a pose loss (not illustrated). Thus, the training module **330** may formulate the generation of the depth map **308** as a photometric error minimization across the images **610** and **620**. The training module **330** can then compare the synthesized image **640** and the original image **610** to determine a monocular loss, for example, the photometric loss **650**. This loss characterizes an accuracy of the depth model **304** in producing the depth map **308**. Thus, the training module **330** can then use the calculated loss to adjust the depth model **304**.

[0092] In the second stage, the network module **320** may function to obtain the depth map **308** for training, for example, as generated during the first stage, retrieved from the data store **205**, or processing the first image **610** according to the depth model **304**. In any case, the training module **330** functions to estimate a single scale **670** for the first image **610** (referred to herein as a first scale or image scale) based on the depth map **308**. For example, the training module functions to generate a point cloud **675** from the depth map **308**. That is, for example, each pixel of the depth map **308** includes X and Y coordinates of the 2D image plane, which can be converted to a 3D point cloud **675** using depth values for each pixel as a Z-coordinate in 3D space. The point cloud **675** may not include points for invalid pixels of the depth map **308**, for example, where the depth values are infinite such as a background of the scene.

[0093] From this point cloud **675**, the training module **330** functions to estimate the first scale **670** for the first image **610**. For example, a distance between points in the point cloud **675** may be measured to recover pixel-wise scale between points. Thus, a scale for each point may be recovered from the point cloud. In some embodiments, the training module **330** may function to obtain a measure of central tendency across the pixel-wise scales to estimate a single first scale **670** for the first image **610**. For example, the training module **330** may calculate an average, median, or mode of the scales for all points as an estimate of the first scale **670** for the first image **610**. In this way, a single scale of a monocular image may be obtained based on the depth values from the depth map **308**. In some embodiments, assuming scale-factor consistency in the image sensor, the first scale **670** may be indicative of a scale corresponding to any image (e.g., image **248**) captured by the image sensor (e.g., image sensor **246**).

[0094] Additionally, the network function **320** functions to obtain depth data **312** for training, for example, from the data store **205** or directly from a range sensor. The depth data **312** is obtained for the same or similar scene as the target image **610** used to calculate a scale value **680** for the depth data (D.sub.t). For example, in various embodiments, depth data is obtained by a range sensor (e.g., radar sensor **242**) and generally includes depth measures and distance information from a point in the scene relative to a point of origin (e.g., such as range sensor). In any case, the training module **330** functions to estimate a single scale **680** (referred to herein as a second scale or range sensor scale) based on the depth data **312**. The depth data **312** may be obtained as a point cloud, where distance information may be represented in X and Y coordinates and the depth measures represented in Z coordinates. The depth data and thus the point cloud **685** includes only those points for which a depth measure is detected by the range sensor. Thus, the point cloud **685** is sparse or generally incomplete for the scene corresponding to image **610**, such that only sparsely distributed points within the scene include the depth measures. Whereas, the depth map that generally provides comprehensive depths for each separate depicted pixel. Furthermore, the depth measures of the depth data **312** are noisy and thus

include a degree of error and uncertainty for each measure.

[0095] Accordingly, from the point cloud **685**, the training module **330** functions to estimate the second scale **680** for the depth data. For example, a distance between points in the point cloud **685** may be measured with depth measures from the radar to recover a point-wise scale for each point in the point cloud **685**. The training module **330** may function to obtain a measure of central tendency across the point-wise scales to estimate a second first scale **680**. For example, the training module **330** calculates an average, weighted average, median, or mode of the scales for all points as an estimate of the single second scale **680**. In one example, a weighted average or median of the point-wise scales of the depth data **312** may be used. For example, a weight may be applied to each scale based on the degree of uncertainty/error in the scales for each pixel and an average taken of the weighted scales (e.g., weights determined by normalizing the uncertainty according to the lowest uncertainty within the depth data **312**). In this way, a single scale for the depth data **312** may be obtained based on the depth measures, which smooths out errors in the depth data and is, therefore, more robust than a pixel-wise application of scales. In some embodiments, assuming scale-factor consistency in the radar sensor, the second scale **680** may be indicative of a scale corresponding to any depth data generated by the range sensor (e.g., radar sensor **242**).

[0096] In some embodiments, once the first and second scales **670** and **680** are determined, the training module **330** functions to generate the supervised loss **660**. For example, the supervised loss **660** can be generated from a comparison of the first scale **670** with the second scale **680**. For example, the second scale **670** may be divided by the first scale **670** to generate a supervised loss **660**. The supervised loss **660** (e.g., the single scale factor) can be used to align the depth model **304** with the second scale **680** associated with the range sensor (e.g., radar sensor **242**). Thus, the training module **330** may use a single scale for the depth data **312** to collapse the scale ambiguity in the target image **610** into a metrically accurate scale factor for use in the depth model **304**; thereby, providing for a scale-aware depth model for monocular depth estimation.

[0097] In another example, the supervised loss **660** may be obtained by dividing the point-wise depth data from point cloud **680** with the pixel-wise distance between points of point cloud **675**. For example, for each valid pixel, the depth values from the depth map **308** may be divided by the depth data (D.sub.t). This pixel-wise comparison provides the conversion from the unknown scale of the target image **610** to the metric scale. The metric scale is maintained using a measure of central tendency, such as a weighted average for the entire dataset. Then, as described above, the supervised loss **660** can be used to align the depth model **304** with metrically accurate scale based on the depth data from range sensor (e.g., radar sensor **242**). Thus, the training module **330** may use a single scale for the depth data **312** to collapse the scale ambiguity in the target image **610** into a metrically accurate scale factor for use in the depth model **304**; thereby, providing for a scale-aware depth model for monocular depth estimation.

[0098] The supervised loss **660** may be used to pay a loss between radar sensor scale and image scale, penalizing this distance. The supervised loss can then be used in addition to the photometric loss **650**, and eventually the supervised loss will be minimized, meaning that the depth data point cloud and image point cloud will have the same scale.

[0099] In some embodiments, the training module **322** may function to determine whether or not the errors/uncertainty in the depth data **312** is above a set threshold uncertainty. Thresholds may be manually set empirically, or using a percentage (e.g., the 50% highest errors in the image are discarded, regardless of the value). For example, the training module **322** may determine a measure of central tendency of the uncertainty in the depth data and compare this to a threshold uncertainty. In another example, the training module **322** identify a number of points in the depth data **312** having uncertainty above the threshold uncertainty and, if that number is greater than a set threshold number, determine that the error/uncertainty in the depth data **322** is too high. Where the training module determines the degree of error/uncertainty is above the threshold, a single scale factor (e.g., supervised loss **660**) is determined from a plurality of readings by the range sensor (e.g., where each reading corresponds to a point cloud such as point cloud **680** above). The plurality of readings may be part of a

dataset, for example, stored in data store **205**. For each reading, a second scale **680** is separately determined as set forth above to generate a plurality of second scales. A measure of central tendency is then taken of the plurality of second scales (e.g., an average, median, mode, etc.) to determine a single aggregate second scale for the depth data. In one embodiment, the supervised loss **660** may be determined as set forth above, except that the single aggregate second scale is used in place of a single second scale.

[0100] In another embodiment, the training module **322** may generate an aggregate first scale from the dataset. For example, the dataset may include the plurality of readings from the range sensor and a plurality of images corresponding to the scenes of the plurality of readings. In this case, for each image, a first scale **670** is separately determined as set forth above to generate a plurality of first scales. A measure of central tendency is then taken of the plurality of first scales (e.g., an average, median, mode, etc.) to determine a single aggregate first scale for the images. The supervised loss **660** may then be determined as set forth above, except that the single aggregate first scale and single aggregate second scale are used in place of a first and second scales.

[0101] In yet another example, a single scale factor may be determined for each training data pair in the data set (e.g., each reading and corresponding image). The training module **330** may then determine a measure of central tendency for the scale factors across the dataset, thereby generating a single aggregate scale factor for the dataset.

[0102] As an example, with reference to self-supervised loss context for SfM, the training module **330** may be configured with (i) a monocular depth model $f_{\text{sub.D:I.fwdarw.D}}$ (e.g., depth model **304**) that predicts a scale-ambiguous depth $\{ \text{grave over (D)} \} = f_{\text{sub.D(I(p))}}$ for every pixel p in the target image $I_{\text{sub.t}}$; and (ii) a monocular motion estimator $f_{\text{sub.X:(I.sub.t, I.sub.s)}}$ (e.g., pose model **306**), that predicts a set of 6-DoF rigid-body transformations for all $s \in S$ given by

$$[00001] \ x_{t-s} = \begin{pmatrix} R & r \\ 0 & 1 \end{pmatrix} \in \text{SE} \quad \text{Eq. 1}$$

[0103] between the target image $I_{\text{sub.t}}$ and the set of source images $I_{\text{sub.s}} \in I_{\text{sub.s}}$ considered as part of a temporal context. Additionally, the training module **330** may use the depth data from the training data **302** to (i) collapse the scale ambiguity of a monocular image sensor configuration into a metrically accurate version of the depth model **304**, and (ii) improve the depth model **304** and the pose model **306** by leveraging cues from the depth data that are not appearance-based.

[0104] The training module **330** may implement the training of the depth model **304** according to two components. The two components include a self-supervised term (e.g., photometric loss **650**) that operates on appearance matching $L_{\text{sub.p}}$ between the target image $I_{\text{sub.t}}$ (e.g., image **610**) and the synthesized image $I_{\text{sub.s.fwdarw.t}}$ (also annotated as $\hat{I}_{\text{sub.t}}$ **640**) from the context set $S = \{I_{\text{sub.S}}\}_{\text{sub.s}=1}^{\text{sup.S}}$, with masking $M_{\text{sub.s}}$ and depth smoothness $L_{\text{sub.smooth}}$, and a supervised loss value that operates on a comparison between the predicted depth map **308** and the depth data **312** from the training data **250**.

$$[00002] \ L_{(I_t, \hat{I}_t)} = L \odot M_{p^+} + L_{\text{smooth}^+} + L_{\text{supervised}} \quad \text{Eq. (2)}$$

[0105] Where M_p is a binary mask that avoids computing the photometric loss on the pixels that do not have a valid mapping (e.g., pixels from the separate images that do not project onto the target image given the estimated depth), $\lambda_{\text{sub.1}}$ and $\lambda_{\text{sub.2}}$ represent weights for adjusting the loss terms in Eq (2). $L_{\text{sub.p}}$ represents appearance matching loss and is implemented according to, a pixel-level similarity between the target image $I_{\text{sub.t}}$ (e.g., image **610**) and the synthesized image $\hat{I}_{\text{sub.t}}$ (e.g., image **640**) using a structural similarity (SSIM) term combined with an L1 pixel-wise loss term inducing an overall photometric loss as shown in:

$$[00003] \ L_p(I_t, \hat{I}_t) = \frac{1 - \text{SSIM}(I_t, \hat{I}_t)}{2} + (1 - \lambda) \cdot \text{Math. } I_t - \hat{I}_t \cdot \text{Math.} \quad \text{Eq. (3)}$$

[0106] While multi-view projective geometry provides strong cues for self-supervision, errors due to parallax and out-of-bounds objects have an undesirable effect incurred on the photometric loss that can include added noise to the training. Accordingly, the training module **230** can mitigate these effects by calculating the minimum photometric loss per pixel for the source image according to:

$$[00004] L_p(I_t, S) = \min_{s \in S} L_p(I_t, I_{s.fwdarw. t}) \quad \text{Eq. (4)}$$

[0107] The intuition involves the same pixel not occluding or be out-of-bounds in all context images, and that the association with minimal photometric loss should be the correct. Additionally, as shown below, the training module **330** masks out static pixels by removing pixels that have a warped photometric loss higher than a corresponding unwarped photometric loss, which the training module **330** calculates using the original source image (e.g., **620**) without synthesizing the target. The mask (Mp) removes pixels that have appearance loss that does not change between frames, which includes static scenes and dynamic objects moving at a similar speed as the image sensor.

$$[00005] M_p = \min_{s \in S} L_p(I_t, I_s) > \min_{s \in S} L_p(I_t, I_{s.fwdarw. t}) \quad \text{Eq. (5)}$$

[0108] L.sub.smooth represents depth smoothness loss and is implemented to regularize the depth in textureless low-image gradient regions, as shown in equation (5) below. The smoothness loss is an edge-aware term that is weighted for separate pyramid levels starting from 1 and decaying by a factor of two for the separate scales.

[00006]

$$L_{\text{smooth}}(\hat{D}_t) = \text{Math.} \quad {}_x \hat{D}_t \cdot \text{Math.} \quad e^{-\text{Math.} \quad {}_x I_t \cdot \text{Math.}} + \text{Math.} \quad {}_y \hat{D}_t \cdot \text{Math.} \quad e^{-\text{Math.} \quad {}_y I_t \cdot \text{Math.}} \quad \text{Eq. (6)}$$

[0109] Thus, the training module **330** calculates the appearance-based loss according to the above to include the photometric loss, the mask, and the depth smoothness terms for the self-supervised first stage. Through this first training stage, the model **304** develops a learned prior of the monocular images as embodied by the internal parameters of the model **304** from the training on the image pairs in the training data **250**. In general, the model **304** develops the learned understanding about how depth relates to various aspects of an image according to, for example, size, perspective, and so on. However, after the first stage of training, the depth model **304** still lacks awareness of a metrically accurate scale.

[0110] Consequently, the training module **330** controls the network module **320** to execute the second stage that is a supervised training process in combination with the self-supervised process as previously described. Thus, the training module **330** further employs the second stage loss (e.g., supervised loss **660**) to refine the depth model **304**. To generate the second stage loss, the training module **330** generates a scale factor as the supervised loss **660** by taking a central tendency of a scales from the depth map **308** of the target image I.sub.t and a central tendency of scales from depth data **312**. That is, for example, pixel-wise scales are calculated from a point cloud of the depth map **304** and the central tendency of the pixel-wise scales is determined. For example, the training module **330** takes an average, median, or mode of the pixel-wise scales to determine a single first scale.

Additionally, point-wise scales are determined from the point cloud of the depth data **312** and the central tendency of the point-wise scales is determined. For example, the point-wise scale value annotated to each valid point the depth data **312** is weighed based on the uncertainty of each depth measurement and then an average may be calculated from the weighted scale values. Thus, the supervised loss **660** is then calculated by taking the ratio of the second scale of the depth data **312** to the first scale of the depth map **308**.

[0111] The training module **330** imposes the supervised loss **660** to further refine the depth model **304**. The additional supervised loss (L.sub.supervised) allows the depth model **304** to learn metrically accurate estimates resulting in the depth model **304** improving predictions. By generating representative losses across the depth data **312**, uncertainty due to noise and sparsity in depth data may be smoothed and reduced. Thereby, providing an reasonable estimate of the scale in the depth data **312**, which can be used to align the scale in a target image with the scale in the depth data. In this way, the semi-supervised learning circuit **210** improves the understanding of the depth model **304** while avoiding a need for denser and more costly annotated depth data to provide scale-aware depth estimates.

[0112] The training module **330** may then train the depth model **304** and the pose model **306** together in an iterative manner over the training data **302** that includes a plurality of monocular images from

process and a plurality of monocular video images with corresponding depth data. Through the process of training the model **304**, the training module **330** adjusts various hyper-parameters in the model **304** to fine-tune the functional blocks included therein. Through this training process, the model **304** develops a learned prior of the monocular images as embodied by the internal parameters of the model **304**. In general, the model **304** develops the learned understanding about how depth relates to various aspects of an image according to, for example, size, perspective, and so on. Consequently, the network module **320** can provide the resulting trained depth model **304** in the semi-supervised learning circuit **210** to estimate depths from monocular images that do not include an explicit modality identifying the depths. In further aspects, the network module **320** may provide the depth model **304** to other systems that are remote from the semi-supervised learning circuit **210** once trained to perform similar tasks. In this way, the semi-supervised learning circuit **210** functions to improve the accuracy of the depth model **304** while using minimal annotated data in a supervising refinement stage of training as opposed to relying on comprehensive training data sets that include extensively annotated data.

[0113] In some embodiments, the supervised loss may be aggregated over an entire dataset (e.g., the plurality of target images and corresponding range sensor reading). The supervised loss in this manner may be used to refine the depth model **304** and allows the depth model **304** to learn metrically accurate estimates; thereby, resulting in the depth model **304** and improving predictions. For example, a supervised loss **660** (e.g., scale factor) can be calculated as set forth above for each target image and range sensor reading pair, thereby generate a plurality of first scales. An aggregate of the supervised losses can be generated by taking a measure of central tendency of all supervised losses **660** calculated for the entire dataset, thereby providing a single supervised loss that can be used to train the depth model **304**. As another example, a first scale **670** may be calculated for each target image and a second scale **680** may be calculated for each reading corresponding to each target image, as set forth above. A measure of central tendency of the first scales, second scales, or both may be taken across the entire dataset, for example, by averaging the first scales **670** (referred to herein as aggregate first scale) and/or averaging the second scales **680** (referred to herein as aggregate second scales). Then, similar to the operations set forth above, a ratio of the aggregate depth data loss to the aggregate photometric loss provides the supervised loss **660**.

[0114] While embodiments herein are provided where the depth data is radar data, as set forth above, the depth data may be from any other suitable range sensor. For example, point clouds generated by LiDAR sensors may be used as depth data.

[0115] FIG. 7 is an operational flow diagram illustrating an example process for semi-supervised scale-aware learning of a depth model for monocular depth estimation in accordance with various embodiments. FIG. 7 illustrates a process **700** that may be implemented as instructions, for example, stored on memory **208**, that when executed by one or more processors **206** to perform the operations of process **700**.

[0116] In some embodiments, training the depth model **304** according to the first stage that is self-supervised may be represented as blocks **710**, **720**, **730**, and **760**. The second stage that is supervised may be represented at blocks **710**, **720**, **730**, **740**, **750**, and **760**. Thus, the second stage may encompass the first stage in addition to further aspects corresponding with blocks **740** and **750**. Thus, training according to the separate stages will generally be described according to method **700** overall. However, it should be appreciated that training according to the separate stages generally may occur independently and over multiple iterations with the first stage executing over a plurality of iterations prior to training according to the second stage over a plurality of separate iterations occurring after the first stage. Once the depth model **304** is trained overall according to both stages, the depth model **304** provides a metrically accurate mechanism for inferring depths from monocular images, which the network module **320** may then provide to another device or use within the existing device to perceive information about an environment, for example, the vehicle **10**.

[0117] At block **710**, training data is received including a pair of monocular training images. In the case of the second stage, the training data also includes depth data. The training data may be, for example, training data **302** of FIG. 3. According to various embodiments, the training module **330**

acquires training images locally from co-located systems with the semi-supervised learning circuit **210** (e.g., image sensor **246**) in an active manner along with the depth data (e.g., distance measuring sensors such as radar **242**). In further embodiments, the training module **330** may acquire the training images and depth data through a communication link with a remote system or from a repository of such information as included in the data store **205**. Thus, while the semi-supervised learning circuit **210** can be implemented within a particular device that is, for example, actively navigating through an environment (e.g., vehicle **10**), the semi-supervised learning circuit **210** may also function as a cloud-based computing service to train the depth model **304** and/or to analyze monocular images for depth information, and thus may receive the training data **250** from separate sources.

[0118] Block **710** may also include receiving depth data associated with the monocular video in order to facilitate the supervised training stage. That is, whether the depth data is embedded with the monocular training images or is provided separately, the training data **302** includes the depth data for, in one approach, at least a subset of the image from the video. As previously mentioned, the depth data is noisy depth data (e.g., derived from a radar sensor **242**) that correlates with the video from image sensor **246**. Thus, the image sensor **246** and a radar sensor **242** may be generally calibrated together in order to correlate the depth data with the images (e.g., for example based on timestamps or other association of data).

[0119] At block **720**, the depth model and the pose model are executed on the pair of training images received at block **710**. For example, the training module **230** causes the network module **220** to execute the depth model **304** and the pose model **306** on the training images from training data **205**. In some embodiments, the network module **320** executes the depth model **304** to process a first image of a training pair to generate the depth map **308**. Execution of the depth model **304** to produce the depth map **308** at block **620** may be a routine execution of the depth model **304**. For example, there is minimal variation in the way in which the model **304** is executed during training since the overall goal is to have the model **304** produce the depth map **308** according to learned weights and as would occur in normal operating conditions so that the training module **330** can subsequently assess the performance of the model **304** according to the loss functions.

[0120] Also at block **720**, the pose model **306** may be executed to process the first image and a second image of the training pair to generate the transformation **310**. The execution of the pose model **306** is generally routine occurs under normal operating conditions. Although the pose model **306** may be used in various circumstances for generating transformations, the semi-supervised learning circuit **210** may use the pose model **306** for the application of training the depth model **304**. Thus, when implemented as part of process **700**, the pose model **306** may reside in an idle state when not training the depth model **304**. Models **304** and **306** may be executed in parallel during training to generate the depth map **308** and the transformation **310**. That is, the pose model **306** may function to facilitate the self-supervised structure from motion (SfM) training by providing the transformation **310** from which the training module **330** may assess the performance of the depth model **304**. The training module **330** may also train the pose model **306**, which may be done in combination with the depth model **304**. Execution of the depth model **304** and the pose model **306** for the first and the second stage generally may occur in the same manner for both stages. However, in the second stage at least the first image further includes the corresponding depth data for subsequent use in generating the second stage loss values as discussed below.

[0121] At block **730**, self-supervised loss(es) (e.g., photometric loss, depth smoothness loss, etc.) are calculated as either independent losses for the first stage or as one component of a second stage loss. For example, the training module **230** may compute the self-supervised loss(s), as described above in connection with FIG. **6**. In some embodiments, the training module **330** may initially calculate the self-supervised loss according to a comparison between a synthetic image and the target image (i.e., the first image of the pair) according to the photometric loss function, which may include an appearance loss, a regularization/depth smoothness loss, and/or other components that are appearance-based.

[0122] At block **740**, a determination is made whether the current training stage is the second. If the

determination is no, process **700** proceeds to block **760** for adjusting the depth model **304**. Otherwise, process **700** proceeds to block **750** for generating the additional second-stage supervised loss, for example, as described above in connection with FIG. **6**. As previously noted, the first stage of training is a self-supervised structure from motion (SfM) training process that accounts for motion of an image sensor between the training images of a pair to cause the depth model to learn how to infer depths without using annotated training data (e.g., without the depth data). However, because the resulting depth model **304** from solely training on the self-supervised process does not accurately understand scale (e.g., is scale ambiguous), the training module **330** further imposes the second stage to refine the depth model **304**. That is, during the second stage, the training module **330** trains the depth model **304** to refine the model **304** using a single scale factor derived from depth data that includes annotations about depth in the individual images. As previously noted, the depth data may represent dispersed ground truths providing limited supervision over depth estimates of the individual images.

[0123] At block **760**, the depth model **304** is updated according to the loss values (e.g., first stage or first/second stage) as determined in blocks **730** and/or **750**. For example, the training module **330** updates the depth model **304** using the loss values to adapt weights in the model **304**. Therefore, the disclosed two-stage semi-supervised training approach improves the understanding of the depth model **304** while using primarily self-supervised training that is supplemented with the additional refinement stage that uses a minimal set of sparsely annotated depth data. In this way, the embodiments herein improve the depth model **304** to produce improved depth estimates that translate into improved situational awareness of the implementing device (e.g., the vehicle **10**), and improved abilities to navigate and perform other functions therefrom.

[0124] FIG. **8** is a flow chart illustrating example operations for implementing semi-supervised scale-aware learning of a depth model for monocular depth estimation in accordance with various embodiments. In this example, the process may be performed by various devices described herein, including the system **200** of FIG. **2**, cloud-based servers, physical or dedicated server, software-as-a-service (SaaS), distributed computing service, etc.

[0125] At block **805**, the process may receive an image captured by an image sensor, the image comprising pixels representing a scene of an environment. For example, the image may be a monocular image captured by a monocular camera.

[0126] At block **810**, the process may derive a depth map for the image based on a depth model. The depth map may include predicted depth values for a plurality of the pixels of the image.

[0127] At block **815**, the process may estimate a first scale for the image based the plurality of predicted depth values. For example, a point cloud may be generated from the depth map based on the predicted values and the first scale estimated from the point cloud. The first scale may be determined by a measure of central tendency of the predicted depth values. The first scale is a single scale for the image, and may be indicative of a scale associated with the image sensor used to capture the image.

[0128] At block **820**, the process receive depth data captured by a range sensor. The depth data may include a point cloud representing the scene of the environment, where the point cloud includes depth measures for a plurality of points of the point cloud. For example, the range sensor may be a radar sensor that provides a sparse point cloud.

[0129] At block **825**, the process may determine a second scale for the point cloud based on the depth measures from block **820**. The second scale may be determined by a measure of central tendency of the depth measures for each point of the point, for example, a weighted average with weights being based on errors associated with each point of the point cloud. The second scale is a single scale for the range sensor, and be indicative of a scale associated with the range sensor used to generate the depth data.

[0130] At block **830**, the process may determine a scale factor based on a comparison of the second scale with the first scale. For example, a scale factor may be determined by dividing the second scale factor by the first scale factor. The scale factor may be a single scale factor providing for scale consistency in the depth model.

[0131] At block **835**, the process may update the depth model based on the scale factor, such that the

depth model generates metrically accurate depth estimates based on the scale factor from block **830**.

[0132] As used herein, the terms circuit and component might describe a given unit of functionality that can be performed in accordance with one or more embodiments of the present application. As used herein, a component might be implemented utilizing any form of hardware, software, or a combination thereof. For example, one or more processors, controllers, ASICs, PLAS, PALs, CPLDs, FPGAs, logical components, software routines or other mechanisms might be implemented to make up a component. Various components described herein may be implemented as discrete components or described functions and features can be shared in part or in total among one or more components. In other words, as would be apparent to one of ordinary skill in the art after reading this description, the various features and functionality described herein may be implemented in any given application. They can be implemented in one or more separate or shared components in various combinations and permutations. Although various features or functional elements may be individually described or claimed as separate components, it should be understood that these features/functionality can be shared among one or more common software and hardware elements. Such a description shall not require or imply that separate hardware or software components are used to implement such features or functionality.

[0133] Where components are implemented in whole or in part using software, these software elements can be implemented to operate with a computing or processing component capable of carrying out the functionality described with respect thereto. One such example computing component is shown in FIG. **9**. Various embodiments are described in terms of this example-computing component **900**. After reading this description, it will become apparent to a person skilled in the relevant art how to implement the application using other computing components or architectures.

[0134] Referring now to FIG. **9**, computing component **900** may represent, for example, computing or processing capabilities found within a self-adjusting display, desktop, laptop, notebook, and tablet computers. They may be found in hand-held computing devices (tablets, PDA's, smart phones, cell phones, palmtops, etc.). They may be found in workstations or other devices with displays, servers, or any other type of special-purpose or general-purpose computing devices as may be desirable or appropriate for a given application or environment. Computing component **900** might also represent computing capabilities embedded within or otherwise available to a given device. For example, a computing component might be found in other electronic devices such as, for example, portable computing devices, and other electronic devices that might include some form of processing capability.

[0135] Computing component **900** might include, for example, one or more processors, controllers, control components, or other processing devices. This can include a processor, and/or any one or more of the components making up user vehicle **10**, semi-supervised scale-aware learning system **200**, and cloud. Processor **904** might be implemented using a general-purpose or special-purpose processing engine such as, for example, a microprocessor, controller, or other control logic. Processor **904** may be connected to a bus **902**. However, any communication medium can be used to facilitate interaction with other components of computing component **900** or to communicate externally.

[0136] Computing component **900** might also include one or more memory components, simply referred to herein as main memory **908**. For example, random access memory (RAM) or other dynamic memory, might be used for storing information and instructions to be executed by processor **904**. Main memory **908** might also be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **904**. Computing component **900** might likewise include a read only memory ("ROM") or other static storage device coupled to bus **902** for storing static information and instructions for processor **904**.

[0137] The computing component **900** might also include one or more various forms of information storage mechanism **910**, which might include, for example, a media drive **912** and a storage unit interface **920**. The media drive **912** might include a drive or other mechanism to support fixed or removable storage media **914**. For example, a hard disk drive, a solid-state drive, a magnetic tape drive, an optical drive, a compact disc (CD) or digital video disc (DVD) drive (R or RW), or other

removable or fixed media drive might be provided. Storage media **914** might include, for example, a hard disk, an integrated circuit assembly, magnetic tape, cartridge, optical disk, a CD or DVD. Storage media **914** may be any other fixed or removable medium that is read by, written to or accessed by media drive **912**. As these examples illustrate, the storage media **914** can include a computer usable storage medium having stored therein computer software or data.

[0138] In alternative embodiments, information storage mechanism **910** might include other similar instrumentalities for allowing computer programs or other instructions or data to be loaded into computing component **900**. Such instrumentalities might include, for example, a fixed or removable storage unit **922** and an interface **920**. Examples of such storage units **922** and interfaces **920** can include a program cartridge and cartridge interface, a removable memory (for example, a flash memory or other removable memory component) and memory slot. Other examples may include a PCMCIA slot and card, and other fixed or removable storage units **922** and interfaces **920** that allow software and data to be transferred from storage unit **922** to computing component **900**.

[0139] Computing component **900** might also include a communications interface **924**. Communications interface **924** might be used to allow software and data to be transferred between computing component **900** and external devices. Examples of communications interface **924** might include a modem or softmodem, a network interface (such as Ethernet, network interface card, IEEE 802.XX or other interface). Other examples include a communications port (such as for example, a USB port, IR port, RS232 port Bluetooth® interface, or other port), or other communications interface. Software/data transferred via communications interface **924** may be carried on signals, which can be electronic, electromagnetic (which includes optical) or other signals capable of being exchanged by a given communications interface **924**. These signals might be provided to communications interface **924** via a channel **928**. Channel **928** might carry signals and might be implemented using a wired or wireless communication medium. Some examples of a channel might include a phone line, a cellular link, an RF link, an optical link, a network interface, a local or wide area network, and other wired or wireless communications channels.

[0140] In this document, the terms “computer program medium” and “computer usable medium” are used to generally refer to transitory or non-transitory media. Such media may be, e.g., memory **908**, storage unit **920**, media **914**, and channel **928**. These and other various forms of computer program media or computer usable media may be involved in carrying one or more sequences of one or more instructions to a processing device for execution. Such instructions embodied on the medium, are generally referred to as “computer program code” or a “computer program product” (which may be grouped in the form of computer programs or other groupings). When executed, such instructions might enable the computing component **900** to perform features or functions of the present application as discussed herein.

[0141] It should be understood that the various features, aspects and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described. Instead, they can be applied, alone or in various combinations, to one or more other embodiments, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the present application should not be limited by any of the above-described exemplary embodiments.

[0142] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. As examples of the foregoing, the term “including” should be read as meaning “including, without limitation” or the like. The term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof. The terms “a” or “an” should be read as meaning “at least one,” “one or more” or the like; and adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known.” Terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time. Instead, they should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. Where this document refers to technologies that would be apparent or known to one of

ordinary skill in the art, such technologies encompass those apparent or known to the skilled artisan now or at any time in the future.

[0143] The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent. The use of the term “component” does not imply that the aspects or functionality described or claimed as part of the component are all configured in a common package. Indeed, any or all of the various aspects of a component, whether control logic or other components, can be combined in a single package or separately maintained and can further be distributed in multiple groupings or packages or across multiple locations.

[0144] Additionally, the various embodiments set forth herein are described in terms of exemplary block diagrams, flow charts and other illustrations. As will become apparent to one of ordinary skill in the art after reading this document, the illustrated embodiments and their various alternatives can be implemented without confinement to the illustrated examples. For example, block diagrams and their accompanying description should not be construed as mandating a particular architecture or configuration.

Claims

1. A method for depth estimation, comprising: deriving a depth map for an image based on a depth model, the depth map comprising a plurality of predicted depth values for a plurality of pixels of the image; estimating a first scale comprising a pixel distance between depth values for the image based the plurality of predicted depth values; receiving depth measures corresponding to an environment in the image; determining a second scale comprising a measure of central tendency for the first scale based on the depth measures, wherein the measure of central tendency, for the first scale, comprises a measure of a central value for probability distribution of the first scale based on the depth measure; determining a scale factor based on a ratio between the second scale and the first scale; and updating the depth model based on the scale factor.
2. The method of claim 1, wherein the image is a monocular image.
3. The method of claim 1 wherein a range sensor produces a sparse point cloud containing the depth measures, wherein the depth measure for each point comprises an error.
4. The method of claim 3, further comprising: determining that the error is greater than a threshold; in response to the determination that the error is greater than the threshold, determining a plurality of scale factors for a plurality of images captured by a image sensor and a plurality of depth data captured by a radar sensor; and determining an aggregate scale factor by aggregating the plurality of scale factors, wherein updating the depth model is based on the aggregate scale factor.
5. The method of claim 3, wherein the range sensor is a radar sensor.
6. The method of claim 1, further comprising: generating a point cloud from the depth map, the point cloud comprises points based on the depth values for the plurality of pixels, wherein the estimated first scale is based on the point cloud of the depth map.
7. The method of claim 1, wherein the first scale is a single first scale for the image.
8. The method of claim 6, further comprising: determining a pixel-wise scale for each of the plurality of the pixels of the image based on a depth value of each respective pixel; and determining a measure of central tendency of the pixel-wise scales, wherein the first scale is estimated based on the determined measure of central tendency, wherein the central tendency indicates an arithmetic average for probability distribution.
9. The method of claim 1, wherein the second scale is a single second scale for the depth measures.
10. The method of claim 8, further comprising: determining a point-wise scale for a plurality of points of a point cloud associated with the depth measures based on a depth measure of each respective point; and determining a measure of central tendency of the point-wise scales, wherein the second scale is estimated based on the determined measure of central tendency, wherein the central tendency

indicates a weighted average for probability distribution.

11. A system, comprising: a memory; and one or more processors that are configured to execute machine readable instructions stored in the memory for performing a method comprising: training a depth model at a first stage according to self-supervised photometric losses generated from at least a first monocular image; determining a single scale factor comprising a ratio between a pixel distance between depth values from a depth map of the first monocular image and a measure of central tendency based on the pixel distance from a sparse point cloud generated by a range sensor, wherein the measure of central tendency comprises a measure of a central value for probability distribution across pixel-wise scales; and training the depth model at a second stage according to a supervised loss based on the single scale factor.

12. The system of claim 11, wherein the range sensor produces a sparse point cloud, wherein the measure of central tendency for each point comprises an error.

13. The system of claim 12, wherein the range sensor is a radar sensor.

14. The system of claim 11, wherein the method further comprises: generating a point cloud from the first monocular image, the point cloud comprises points based on the depth values for a plurality of pixels derived from the depth model, wherein the depth map is based on the point cloud of the first monocular image.

15. The system of claim 11, wherein determining the single scale factor from the depth map of the first monocular image and the sparse point cloud generated by the range sensor comprises: estimating a first scale for the first monocular image based on predicted depth values from the depth map, the first scale being a single first scale for the image.

16. The system of claim 15, wherein the method further comprises: determining the pixel-wise scales for the first monocular image based on a depth value of each pixel of the first monocular image; and determining a measure of central tendency of the pixel-wise scales, wherein the first scale is estimated based on the determined measure of central tendency.

17. The system of claim 15, wherein determining the single scale factor from the depth map of the first monocular image and the sparse point cloud generated by the range sensor comprises: receiving depth data captured by a range sensor, the depth data comprising the sparse point cloud representing a scene of an environment of the first monocular image, the point cloud comprising depth measures for a plurality of points of the point cloud; and determining a second scale for the point cloud based on the depth measures, the second scale is a single second scale for the depth data.

18. The system of claim 17, wherein the method further comprises: determining a point-wise scale for each of the plurality of the points of the point cloud based on a depth measure of each respective point; and determining a measure of central tendency of the point-wise scales, wherein the second scale is estimated based on the determined measure of central tendency.

19. The system of claim 17, wherein the depth measure for each point comprises an error, the method further comprising: determining that the error is greater than a threshold; and in response to the determination that the error is greater than the threshold, determining a plurality of scale factors for a plurality of images captured by an image sensor and a plurality of depth data captured by a radar sensor; and determining an aggregate scale factor by aggregating the plurality of scale factors, wherein updating the depth model is based on the aggregate scale factor.

20. The system of claim 17, wherein the single scale factor is based on a comparison of the second scale with the first scale.
